INCH-POUND
MIL-PRF-22684/8C
w/Amendment 4
19 April 2022
SUPERSEDING
MIL-PRF-22684/8C
w/Amendment 3
4 July 2020

PERFORMANCE SPECIFICATION SHEET

RESISTOR, FIXED, FILM, INSULATED, STYLE RL42....TX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-22684.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for insulated, film, fixed resistors of 2-percent and 5-percent resistance tolerance. These resistors are capable of full-load operation at an ambient temperature of 70°C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/°C). Designers are CAUTIONED on using these resistors in high power pulse applications (see 6.3).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:

M22684/08	-	1001
Performance specification number		Coded dash number

The coded dash number is derived in accordance with paragraph 6.2.2 of MIL-PRF-22684.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/

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FSC 5905
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DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-22684	-	Resistor, Fixed, Film (Insulated), General Specification.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Test Methods for Electronic and Electrical Components Parts.
MIL-STD-202-108	-	Test Method Standard Method 108, Life (At Elevated Ambient Temperature)
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

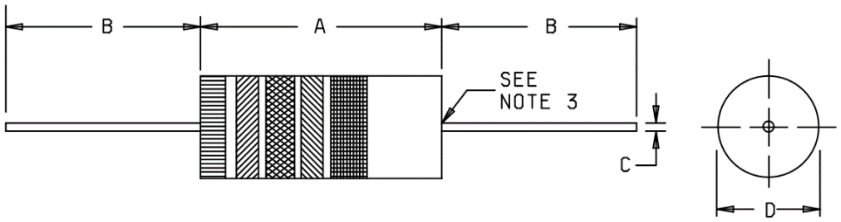
(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-22684.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.



Ltr	DIMENSIONS			
	INCHES		MILLIMETERS	
	Minimum	Maximum	Minimum	Maximum
A	0.648	0.728	16.46	18.49
B	1.375	1.625	34.92	41.28
C	0.040	0.047	1.09	1.19
D	0.280	0.336	7.11	8.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within 0.125 inch (3.18 mm) of the resistor body.

FIGURE 1. Configuration and dimensions.

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3.3 Power rating. The power rating shall be 2 watts based on full load operation at an ambient temperature of 70°C.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 500 volts.

3.5 Resistance values and resistance tolerances. The minimum and maximum standard resistance values and associated resistance tolerances shall be as listed in table I.

Table I. Part Number Designation. (M22684/08-)

Dash No.	Type designation	Resistance tolerance (percent)	Nominal total resistance value (in Ohms)
1001	RL42S100GTX	2	10
1002	RL42S100JTX	5	
1003	RL42S110GTX	2	11
1004	RL42S110JTX	5	
1005	RL42S120GTX	2	12
1006	RL42S120JTX	5	
1007	RL42S130GTX	2	13
1008	RL42S130JTX	5	
1009	RL42S150GTX	2	15
1010	RL42S150JTX	5	
1011	RL42S160GTX	2	16
1012	RL42S160JTX	5	
1013	RL42S180GTX	2	18
1014	RL42S180JTX	5	
1015	RL42S200GTX	2	20
1016	RL42S200JTX	5	
1017	RL42S220GTX	2	22
1018	RL42S220JTX	5	
1019	RL42S240GTX	2	24
1020	RL42S240JTX	5	
1021	RL42S270GTX	2	27
1022	RL42S270JTX	5	
1023	RL42S300GTX	2	30
1024	RL42S300JTX	5	
1025	RL42S330GTX	2	33
1026	RL42S330JTX	5	
1027	RL42S360GTX	2	36
1028	RL42S360JTX	5	
1029	RL42S390GTX	2	39
1030	RL42S390JTX	5	
1031	RL42S430GTX	2	43
1032	RL42S430JTX	5	
1033	RL42S470GTX	2	47
1034	RL42S470JTX	5	
1035	RL42S510GTX	2	51
1036	RL42S510JTX	5	
1037	RL42S560GTX	2	56
1038	RL42S560JTX	5	
1039	RL42S620GTX	2	62
1040	RL42S620JTX	5	
1041	RL42S680GTX	2	68
1042	RL42S680JTX	5	
1043	RL42S750GTX	2	75
1044	RL42S750JTX	5	
1045	RL42S820GTX	2	82
1046	RL42S820JTX	5	
1047	RL42S910GTX	2	91
1048	RL42S910JTX	5	
1049	RL42S101GTX	2	100
1050	RL42S101JTX	5	
1051	RL42S111GTX	2	110
1052	RL42S111JTX	5	
1053	RL42S121GTX	2	120
1054	RL42S121JTX	5	
1055	RL42S131GTX	2	130
1056	RL42S131JTX	5	
1057	RL42S151GTX	2	150
1058	RL42S151JTX	5	
1059	RL42S161GTX	2	160
1060	RL42S161JTX	5	
1061	RL42S181GTX	2	180
1062	RL42S181JTX	5	
1063	RL42S201GTX	2	200
1064	RL42S201JTX	5	
1065	RL42S221GTX	2	220
1066	RL42S221JTX	5	
1067	RL42S241GTX	2	240
1068	RL42S241JTX	5	
1069	RL42S271GTX	2	270
1070	RL42S271JTX	5	
1071	RL42S301GTX	2	300
1072	RL42S301JTX	5	
1073	RL42S331GTX	2	330
1074	RL42S331JTX	5	
1075	RL42S361GTX	2	360
1076	RL42S361JTX	5	
1077	RL42S391GTX	2	390
1078	RL42S391JTX	5	
1079	RL42S431GTX	2	430
1080	RL42S431JTX	5	
1081	RL42S471GTX	2	470
1082	RL42S471JTX	5	
1083	RL42S511GTX	2	510
1084	RL42S511JTX	5	

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Table I. Part number designation (M22684/08-) - continued.

Dash No.	Type designation	Resistance tolerance (percent)	Nominal total resistance value (in Ohms)
1085	RL42S561GTX	2	560
1086	RL42S561JTX	5	
1087	RL42S621GTX	2	620
1088	RL42S621JTX	5	
1089	RL42S681GTX	2	680
1090	RL42S681JTX	5	
1091	RL42S751GTX	2	750
1092	RL42S751JTX	5	
1093	RL42S821GTX	2	820
1094	RL42S821JTX	5	
1095	RL42S911GTX	2	910
1096	RL42S911JTX	5	
1097	RL42S102GTX	2	1,000
1098	RL42S102JTX	5	
1099	RL42S112GTX	2	1,100
1100	RL42S112JTX	5	
1101	RL42S122GTX	2	1,200
1102	RL42S122JTX	5	
1103	RL42S132GTX	2	1,300
1104	RL42S132JTX	5	
1105	RL42S152GTX	2	1,500
1106	RL42S152JTX	5	
1107	RL42S162GTX	2	1,600
1108	RL42S162JTX	5	
1109	RL42S182GTX	2	1,800
1110	RL42S182JTX	5	
1111	RL42S202GTX	2	2,000
1112	RL42S202JTX	5	
1113	RL42S222GTX	2	2,200
1114	RL42S222JTX	5	
1115	RL42S242GTX	2	2,400
1116	RL42S242JTX	5	
1117	RL42S272GTX	2	2,700
1118	RL42S272JTX	5	
1119	RL42S302GTX	2	3,000
1120	RL42S302JTX	5	
1121	RL42S332GTX	2	3,300
1122	RL42S332JTX	5	
1123	RL42S362GTX	2	3,600
1124	RL42S362JTX	5	
1125	RL42S392GTX	2	3,900
1126	RL42S392JTX	5	
1127	RL42S432GTX	2	4,300
1128	RL42S432JTX	5	
1129	RL42S472GTX	2	4,700
1130	RL42S472JTX	5	
1131	RL42S512GTX	2	5,100
1132	RL42S512JTX	5	
1133	RL42S562GTX	2	5,600
1134	RL42S562JTX	5	
1135	RL42S622GTX	2	6,200
1136	RL42S622JTX	5	
1137	RL42S682GTX	2	6,800
1138	RL42S682JTX	5	
1139	RL42S752GTX	2	7,500
1140	RL42S752JTX	5	
1141	RL42S822GTX	2	8,200
1142	RL42S822JTX	5	
1143	RL42S912GTX	2	9,100
1144	RL42S912JTX	5	
1145	RL42S103GTX	2	10,000
1146	RL42S103JTX	5	
1147	RL42S113GTX	2	11,000
1148	RL42S113JTX	5	
1149	RL42S123GTX	2	12,000
1150	RL42S123JTX	5	
1151	RL42S133GTX	2	13,000
1152	RL42S133JTX	5	
1153	RL42S153GTX	2	15,000
1154	RL42S153JTX	5	
1155	RL42S163GTX	2	16,000
1156	RL42S163JTX	5	
1157	RL42S183GTX	2	18,000
1158	RL42S183JTX	5	
1159	RL42S203GTX	2	20,000
1160	RL42S203JTX	5	
1161	RL42S223GTX	2	22,000
1162	RL42S223JTX	5	
1163	RL42S243GTX	2	24,000
1164	RL42S243JTX	5	
1165	RL42S273GTX	2	27,000
1166	RL42S273JTX	5	
1167	RL42S303GTX	2	30,000
1168	RL42S303JTX	5	
1169	RL42S333GTX	2	33,000
1170	RL42S333JTX	5	
1171	RL42S363GTX	2	36,000
1172	RL42S363JTX	5	
1173	RL42S393GTX	2	39,000
1174	RL42S393JTX	5	
1175	RL42S433GTX	2	43,000
1176	RL42S433JTX	5	
1177	RL42S473GTX	2	47,000
1178	RL42S473JTX	5	
1179	RL42S513GTX	2	51,000
1180	RL42S513JTX	5	

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Table I. Part number designation (M22684/08-) - continued.

Dash No.	Type designation	Resistance tolerance (percent)	Nominal total resistance value (in Ohms)
1181	RL42S563GTX	2	56,000
1182	RL42S563JTX	5	
1183	RL42S623GTX	2	62,000
1184	RL42S623JTX	5	
1185	RL42S683GTX	2	68,000
1186	RL42S683JTX	5	
1187	RL42S753GTX	2	75,000
1188	RL42S753JTX	5	
1189	RL42S823GTX	2	82,000
1190	RL42S823JTX	5	
1191	RL42S913GTX	2	91,000
1192	RL42S913JTX	5	
1193	RL42S104GTX	2	100,000
1194	RL42S104JTX	5	
1195	RL42S114GTX	2	110,000
1196	RL42S114JTX	5	
1197	RL42S124GTX	2	120,000
1198	RL42S124JTX	5	
1199	RL42S134GTX	2	130,000
1200	RL42S134JTX	5	
1201	RL42S154GTX	2	150,000
1202	RL42S154JTX	5	
1203	RL42S164GTX	2	160,000
1204	RL42S164JTX	5	
1205	RL42S184GTX	2	180,000
1206	RL42S184JTX	5	
1207	RL42S204GTX	2	200,000
1208	RL42S204JTX	5	
1209	RL42S224GTX	2	220,000
1210	RL42S224JTX	5	
1211	RL42S244GTX	2	240,000
1212	RL42S244JTX	5	
1213	RL42S274GTX	2	270,000
1214	RL42S274JTX	5	
1215	RL42S304GTX	2	300,000
1216	RL42S304JTX	5	

Dash No.	Type designation	Resistance tolerance (percent)	Nominal total resistance value (in Ohms)
1217	RL42S334GTX	2	330,000
1218	RL42S334JTX	5	
1219	RL42S364GTX	2	360,000
1220	RL42S364JTX	5	
1221	RL42S394GTX	2	390,000
1222	RL42S394JTX	5	
1223	RL42S434GTX	2	430,000
1224	RL42S434JTX	5	
1225	RL42S474GTX	2	470,000
1226	RL42S474JTX	5	
1227	RL42S514GTX	2	510,000
1228	RL42S514JTX	5	
1229	RL42S564GTX	2	560,000
1230	RL42S564JTX	5	
1231	RL42S624GTX	2	620,000
1232	RL42S624JTX	5	
1233	RL42S684GTX	2	680,000
1234	RL42S684JTX	5	
1235	RL42S754GTX	2	750,000
1236	RL42S754JTX	5	
1237	RL42S824GTX	2	820,000
1238	RL42S824JTX	5	
1239	RL42S914GTX	2	910,000
1240	RL42S914JTX	5	
1241	RL42S105GTX	2	1,000,000
1242	RL42S105JTX	5	
1243	RL42S115GTX	2	1,100,000
1244	RL42S115JTX	5	
1245	RL42S125GTX	2	1,200,000
1246	RL42S125JTX	5	
1247	RL42S135GTX	2	1,300,000
1248	RL42S135JTX	5	
1249	RL42S155GTX	2	1,500,000
1250	RL42S155JTX	5	

3.6 Terminal type. The terminal type available shall be solderable in accordance with MIL-PRF-22684.

3.7 Thermal shock. Resistors shall be tested as specified in MIL-PRF-22684 except 150°C +3°C, -0°C shall be used instead of 85°C +3°C, -0°C.

3.8 Dielectric withstanding voltage. Resistors shall be tested as specified in MIL-PRF-22684. The magnitude of test voltage shall be as follows:

Atmospheric pressure - 1,000 volts rms
Barometric pressure - 500 volts rms

3.9 Insulation resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the insulation resistance shall be not less than 100 megohms.

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3.10 Moisture resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 2.0 percent.

3.11 Life. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 3.0 percent.

3.12 Power conditioning. When resistors are tested as specified in 4.2, there shall be no mechanical damage.

3.13 Marking. At the option of the manufacturer, parts may be marked in accordance with method I of MIL-STD-1285. The PIN, FSCM, date and lot code, resistance value, and tolerance shall be marked on the parts as in the following example:

M22684/ 08-1239	-	PIN
12345	-	Source code
8401AA	-	Date and lot code
910K 2%	-	Resistance value and tolerance

If this type marking is used, then the green colored band for TX identification may be omitted. Marking shall remain legible after all tests. Full marking required on the unit package.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with Group A inspection and Group B inspection of MIL-PRF-22684 except power conditioning shall be performed on 100 percent of the product supplied under this specification.

4.2 Power conditioning (see 3.12). Resistors shall be tested in accordance with MIL-STD-202-108. The following details and exceptions shall apply:

a. Measurement before conditioning: DC resistance measured in accordance with paragraph 3.7 of MIL-PRF-22684.
b. Conditioning procedures: Each resistor supplied under this specification shall be subjected to power conditioning for a period of 24 hours at 150 percent rated power. Mounting shall be such that 25°C, +10°C, -5°C ambient air can be circulated around the resistors by means of forced air with a velocity not exceeding 500 FPM. The maximum applied voltage shall not exceed 500 V rms.
c. Measurement after conditioning: After power conditioning, the resistors shall be stabilized at room conditions for a minimum of 2 hours before the dc resistance measurements are made. Resistors that are outside of the tolerance limit and those that have changed more than 0.5 percent due to the conditioning shall be removed from the lot. When the combined quantity removed exceeds 10 percent of the lot, the entire lot shall be rejected. The supplier shall have a complete record available of all test results obtained during power conditioning testing (see ordering data).

4.3 TX identification. Resistors procured to, and meeting all of the criteria specified herein shall bear a green colored band in place of the band "E" in accordance with appendix B to MIL-PRF-22684. The symbol "TX" shall be printed at one of the locations specified below:

a. Preferred: On the green colored band described above.
b. On the packing which packaged the items in groups.

In the event the resistor sample fails to meet the requirements of MIL-PRF-22684 and this specification sheet, the manufacturer shall remove the green band from the sample tested and also from all the resistors represented by the sample.

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5. PACKAGING

5. 1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes The notes specified in MIL-PRF-22684 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, and complete PIN (see 1.2).
b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.
c. Packaging requirements.

6.3 Pulse applications. Designers are CAUTIONED on using these resistors in high power pulse applications. Since they have not been qualified nor tested for such applications, damage and premature failure are possible. These resistors only see a one time pulse (Short-time overload) as part of the group B inspection of this specification. Designers MAY CONSIDER using DLA Land and Maritime drawing 03008 for high power pulse applications.

6.4 Amendment notations. The margins of this specification are marked with vertical lines to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, AT, CR4, MI
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5905-2022-016)

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